Are Headtracking Measures an Indicator of Depressive Symptoms?

Mid-Submission Report — VR 360° Experiment

Team: Behavior Nomos
Prakarsh Jain Adarsh Sharma Zia Mahmood Hussain
Project Repository: `https://github.com/Zia-Mahmood/BRSM_Project`

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1 Introduction

1.1 Background

Depression is one of the most prevalent mental health conditions worldwide, affecting approximately 280 million people globally [11]. Among its diverse symptomatology, *psychomotor retardation*—a slowing of physical movement and cognitive processing—has long been recognised as a core feature of depressive disorders [1]. Behavioural manifestations of psychomotor slowing include reduced exploratory activity, diminished engagement with the environment, and decreased overall movement [2].

Virtual Reality (VR) provides an ecologically valid environment for measuring such behaviour because participants interact with immersive environments while their head movements are continuously tracked. Unlike traditional laboratory paradigms that rely on self-report or constrained motor tasks, VR-based headtracking captures naturalistic exploratory behaviour—the degree to which individuals visually scan and orient themselves within an environment [4].

The present study investigates whether VR headtracking data can reveal behavioural indicators of depressive symptoms. Participants experienced immersive 360° videos while wearing a Meta Quest 3 headset, which recorded head orientation (pitch, yaw, roll) and rotational speed. Alongside the VR experience, participants completed standardised psychological questionnaires measuring depression (PHQ-9), anxiety (GAD-7), trait anxiety (STAI-T), and affect (PANAS).

1.2 Research Question

The central research question of this study is:

> **Do depressive symptoms influence head-movement behaviour during VR exploration?**

Understanding this relationship could contribute to the development of digital behavioural markers of mental health that go beyond traditional self-report assessments.

1.3 Hypotheses

We hypothesise that participants with higher PHQ-9 depression scores will exhibit reduced psychomotor activity, reflected in lower mean head rotation speed and reduced variability in head orientation (Primary). Additionally, higher anxiety scores (GAD-7) may be associated with increased or erratic head scanning behaviour, particularly in emotionally intense environments (Secondary). Finally, different VR environments will induce distinct emotional responses that modulate these head-movement patterns (Tertiary).

1.4 Dataset

The dataset comprises high-frequency time-series head-tracking metrics (position and rotational speed) and discrete clinical self-report scores from 40 college students.

2 Methods

2.1 Participants

The sample comprised 40 college students (36 male, 4 female; mean age = 22.78 years, SD = 1.80, range: 20–28). Participants were recruited through convenience sampling and provided informed consent before taking part in the study. Prior VR experience was recorded using a binary scale (1 = No, 2 = Yes); 16 participants reported prior VR experience, while 24 did not. No participants were excluded from analysis (see Section 2.6 for exclusion criteria).

2.2 Experimental Procedure

Participants viewed five 360° VR videos presented through a Meta Quest 3 headset. Videos were selected to span a range of emotional content: V1 (Abandoned buildings; Negative), V2 (Evening beach; Positive), V3 (University campus; Neutral), V4 (Horror/The Nun; High arousal), and V5 (Tahiti surf; Positive). Before the VR experience, participants completed the PHQ-9, GAD-7, STAI-T, and a pre-experiment PANAS. After viewing each video, participants rated the emotional valence (1 = unpleasant, 9 = pleasant) and arousal (1 = calming, 9 = exciting), alongside their sense of immersion. Post-experiment, they completed the PANAS and the VRISE simulator sickness scale. Headset sensors recorded head orientation and rotational speed at approximately 70 Hz.

2.3 Measures and Metrics

Psychological Measures: Participants completed the PHQ-9 (9-item depression screen; scores ≥ 5 indicate mild depression), GAD-7 (7-item anxiety measure), STAI-T (20-item trait anxiety instrument), PANAS (pre/post-VR affect), and VRISE (simulator sickness severity).

Head-Tracking Metrics: Primary metrics extracted from the 70 Hz time-series data included mean rotation speed (°/s) to reflect overall exploration, SD of yaw (°) for left–right scanning variability, and SD of pitch (°) for up–down scanning. Secondary metrics included SD of roll (°) and total movement magnitude. Files with fewer than 10 valid samples were excluded.

2.4 Participant Grouping

Using the clinically validated PHQ-9 cutoff of ≥ 5 to identify at least mild depressive symptomatology, participants were categorised into two balanced groups: Non-Depressed ($n = 20$, PHQ-9 range 0–4) and Depressed ($n = 20$, PHQ-9 range ≥ 5). A secondary cutoff of ≥ 10 ($n = 8$ vs. 32) was established for sensitivity analysis.

2.5 Statistical Analysis

Analyses were conducted in R (v4.5.2). The pipeline began with descriptive statistics, distributional checks (Shapiro-Wilk tests, Q-Q plots), and outlier detection (± 2.5 z-scores or $1.5 \times$IQR). The primary hypothesis was tested by comparing the two depression groups across 25 metric-video combinations using Welch's t-test or the Mann-Whitney U test depending on normality, with Benjamini-Hochberg FDR adjustments applied. Bivariate relationships were assessed using Pearson correlations. Within-subjects video effects were evaluated using Friedman tests, followed by Bonferroni-corrected post-hoc Wilcoxon signed-rank tests and Kruskal-Wallis tests for emotional category comparisons.

2.6 Missing Data and Exclusions

All 40 participants had complete survey data and headtracking recordings for videos V1–V4. Two participants had missing headtracking data for V5 (filenames did not match). Missing data on text-response fields (emotional descriptions, feedback) did not affect quantitative analyses. Three participants (IDs 21, 25, 35) had relatively low VRISE scores (17, 20, and 24, respectively), indicating minimal simulator sickness; their headtracking recordings appeared valid and no sensor artifacts were observed. With only $N = 40$, excluding participants would substantially reduce statistical power. Therefore, all 40 participants were retained for analysis.

2.7 Outlier Inspection

Headtracking variables were screened using standard IQR-based criteria, and the few observations that fell outside the expected range were reviewed. No data were excluded due to recording artefacts.

3 Results

3.1 Descriptive Statistics

3.1.1 Psychological Measures

The sample exhibited a range of depressive symptomatology: mean PHQ-9 = 6.03 (SD = 4.63, median = 4.50, range: 0–18). The distribution was positively skewed, with the majority of participants in the minimal-to-mild range. Mean GAD-7 was 5.00 (SD = 4.31), and mean STAI-T was 45.00 (SD = 14.63), reflecting generally low-to-moderate anxiety. Mean VRISE was 31.88 (SD = 3.98); higher VRISE scores correspond to more severe simulator sickness symptoms, so this indicates that most participants reported low-to-moderate discomfort.

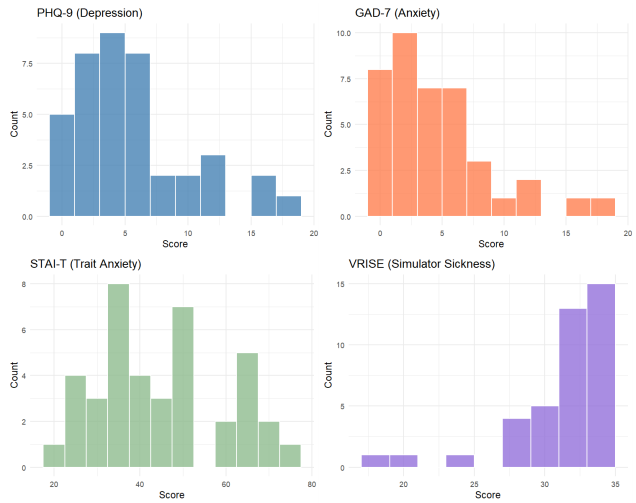

Figure 1: Distributions of psychological scales across 40 participants. PHQ-9 shows a positive skew; most participants fall in the minimal-to-mild depression range.

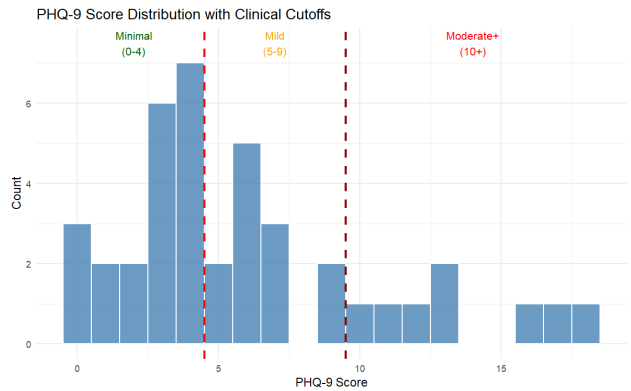

Figure 2: PHQ-9 score distribution with clinical cutoff thresholds. Dashed lines indicate the boundaries for Minimal (0–4), Mild (5–9), and Moderate+ ($\geq$10) categories.

Using the PHQ-9 $\geq$ 5 cutoff produced two balanced groups (Non-Depressed: n = 20, M = 2.60; Depressed: n = 20, M = 9.45). As expected, the depressed group also scored

higher on GAD-7 (M = 7.05 vs. 2.95) and STAI-T (M = 52.25 vs. 37.75), consistent with well-documented comorbidity between depression and anxiety.

3.1.2 Headtracking Measures

Mean rotation speed varied across videos, with V1 (Abandoned) eliciting the highest overall speed ($M = 39.09°/s$, $SD = 10.84$) and V4 (Horror) the lowest ($M = 24.17°/s$, $SD = 11.08$). Table 1 summarises headtracking metrics by video.

Table 1: Headtracking summary statistics by video.

Video	Mean Speed (°/s)	*SD*	SD Yaw (°)	SD Pitch (°)
V1: Abandoned	39.09	10.84	90.0	17.0
V2: Beach	32.76	9.09	88.0	11.5
V3: Campus	34.88	10.48	90.0	13.0
V4: Horror	24.17	11.08	33.0	13.5
V5: Surf	31.16	11.57	85.0	13.5

Shapiro-Wilk tests on the participant-level averages (across videos) indicated that mean speed ($W = 0.977$, $p = .584$) and SD of pitch ($W = 0.973$, $p = .438$) were normally distributed, while SD of yaw was significantly non-normal ($W = 0.914$, $p = .005$). SD of roll was marginally normal ($W = 0.948$, $p = .064$).

3.2 Correlation Analysis

3.2.1 PHQ-9 and Headtracking

A Pearson correlation between PHQ-9 scores and the average mean rotation speed (across all videos) yielded a non-significant result: $r = -.063$, $p = .697$. This indicates that at the aggregate level, there is no meaningful linear relationship between depression severity and overall head-movement speed.

The full correlation matrix (Figure 3) reveals the internal associations among psychological and headtracking variables. PHQ-9 correlated strongly with GAD-7 ($r = .58$, $p < .001$) and STAI-T ($r = .64$, $p < .001$), reflecting the known comorbidity. Pre-experiment negative affect (PANAS-) showed a moderate negative correlation with average speed ($r = -.32$, $p = .042$), suggesting that participants reporting higher negative mood exhibited slightly less head movement.

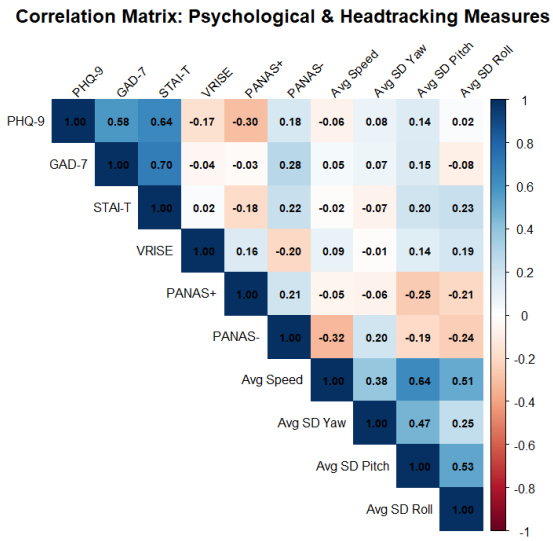

Figure 3: Correlation matrix showing relationships among psychological measures and averaged headtracking metrics. PHQ-9 is strongly associated with GAD-7 and STAI-T, but only weakly related to headtracking measures.

3.3 Group Comparison: Depression and Head Movement

The core hypothesis was tested by comparing the Non-Depressed ($n = 20$) and Depressed ($n = 20$) groups on each headtracking metric for each video (25 tests total). None of the comparisons reached statistical significance at $p < .05$ (uncorrected), and all Benjamini-Hochberg adjusted p-values exceeded .88.

Several medium-sized effects were observed despite the lack of significance:

- **V5 (Surf), Mean Speed:** Non-Depressed $M = 34.35°/\text{s}$ vs. Depressed $M = 28.12°/\text{s}$, Welch's t-test $p = .105$, $d = 0.54$ (medium).
- **V4 (Horror), SD Yaw:** Non-Depressed $M = 37.12°$ vs. Depressed $M = 29.01°$, Mann-Whitney $p = .239$, $d = -0.55$ (medium).
- **V4 (Horror), Total Movement:** Mann-Whitney $p = .120$, $d = -0.54$ (medium).

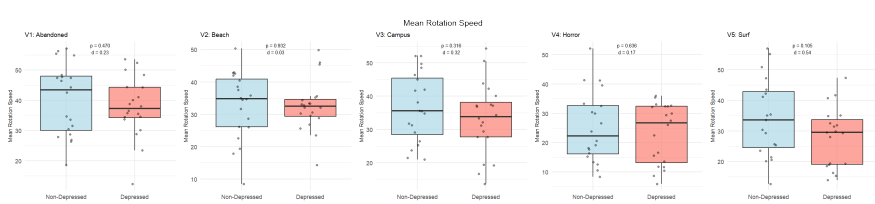

Figure 4: Group comparison for mean rotation speed across five videos. Non-Depressed (blue) vs. Depressed (red) groups. No comparison reached statistical significance.

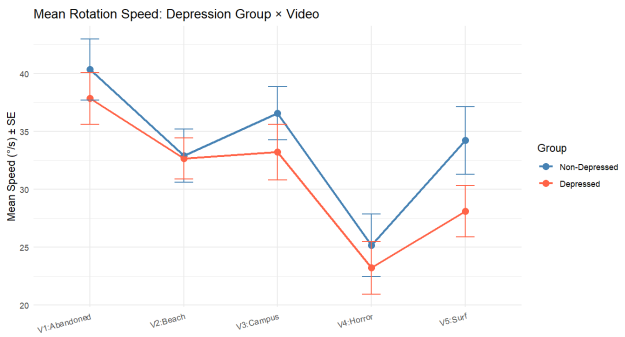

Figure 5: Interaction plot showing mean rotation speed (± SE) for each depression group across the five videos. Both groups show the same pattern of lowest speed during V4 (Horror). The depressed group tends to show lower speed, particularly for V5 (Surf).

3.4 Emotional Response Across Videos

3.4.1 Valence and Arousal Ratings

The five videos elicited clearly differentiated emotional responses (Figure 6). V5 (Surf) anchored the "Excited/Happy" quadrant, V4 (Horror) mapped to the "Tense/Stressed" quadrant, and V1–V3 clustered near the neutral-to-positive midpoint, confirming the successful induction of targeted affective states.

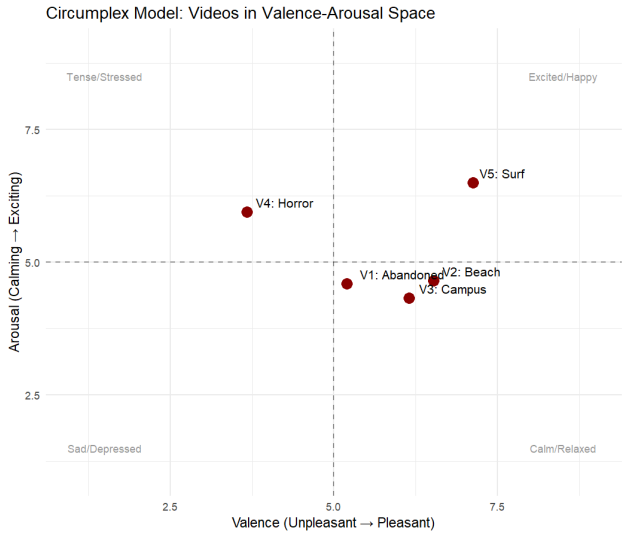

Figure 6: Circumplex model placing each video's mean valence and arousal ratings. V4 (Horror) and V5 (Surf) occupy high-arousal quadrants; V1, V2, and V3 cluster near the neutral midpoint.

3.4.2 Video Effects on Head Movement (Friedman Tests)

Friedman tests revealed highly significant differences in headtracking metrics across the five videos:

- **Mean rotation speed:** $\chi^2(4) = 66.76$, $p < .001$
- **SD of yaw:** $\chi^2(4) = 91.75$, $p < .001$
- **SD of pitch:** $\chi^2(4) = 54.48$, $p < .001$

Post-hoc Wilcoxon signed-rank tests (Bonferroni-corrected) showed that V4 (Horror) differed significantly from all other videos on both mean speed and SD of yaw. This is consistent with the constrained visual focus elicited by the horror content (narrow field of interest, freezing behaviour).

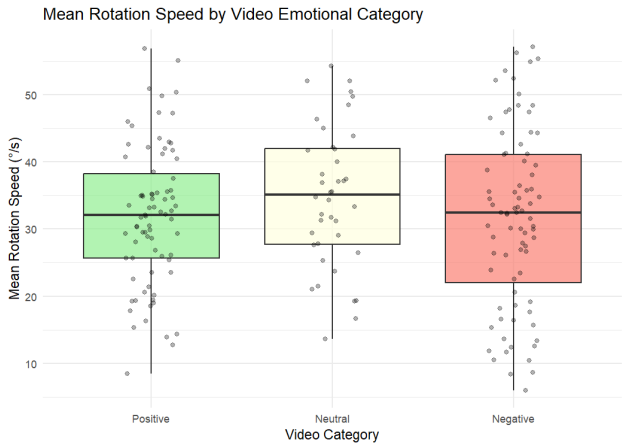

Figure 7: Mean rotation speed by emotional video category. Horror-related videos (Negative) show reduced movement compared to Positive and Neutral content.

4 Conclusion and Future Directions

Our initial analyses indicate no statistically significant group differences between participants with low and elevated depressive symptoms on headtracking metrics. The zero-order correlation between aggregate PHQ-9 scores and mean rotation speed was negligible ($r = -.063$, $p = .697$). However, medium effect sizes ($d \approx 0.5$) during emotionally salient videos (V4 Horror, V5 Surf) suggest that high-arousal contexts may amplify behavioural differences, aligning with theories of emotional context insensitivity in depression (Rottenberg, 2005). Notably, video type exerted a highly significant effect ($p < .001$) on all primary head-movement metrics, confirming that VR content strongly modulates scanning behaviour. The lack of significant group differences is likely limited by the small sample size ($N = 40$, $\approx .37$ power) and a skewed PHQ-9 distribution predominantly reflecting mild rather than clinical depression.

To address these limitations and isolate the unique contribution of depression, our final report will incorporate partial correlations controlling for anxiety (GAD-7, STAI-T), and stratified analyses across anxiety subgroups. We will also analyze pre/post PANAS affective changes, immersion correlations, and conduct sensitivity analyses using the stricter PHQ-9 ≥ 10 clinical cutoff alongside rigorous multiple comparison corrections.

5 Author Contributions

The project workload was divided equally among the team: **Prakarsh Jain** led the experimental design and developed the data preprocessing scripts; **Adarsh Sharma** conducted the core statistical analyses and generated the data visualizations; and **Zia Mahmood Hussain** drafted the manuscript, synthesised the literature, and managed the final report formatting.

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